

BRAC UNIVERSITY
Department of Computer Science and Engineering

CSE 260: Digital Logic Design
Md. Arifur Rahman [MAR]

Assessment: **Assignment 2**
Deadline: **31st October, 2022**

Semester: **Fall 2022**
Full Marks: **10**

*This assignment must be hand-written. Show **all** steps in **all** questions.*

0. Perform the following arithmetic operations using 8-bit 2's complement method. State/Justify if there is an overflow:

- a. Add +91 with -325
- b. Subtract -34 from +67

1. **Simplify** the following Boolean expression to minimum number of literals using the laws of Boolean algebra:

- a. $x'yz' + x'yz + xyz + x'yz'$
- b. $(x' + y')(x + y)$

NB: You cannot use a truth table or K-map to solve this problem.

2. Find the complement of the following expression:

- a. $x'y' + xy'$
- b. $(x' + y + z')(x' + y')(x + z')$

3. Draw the following functions using NAND gates only:

- a. $F(A,B,C) = A' + A'C$
- b. $F(A,B,C,D) = (A'B'CD' + A'D + (B+D'))$
- c. $F(a,b,c) = \prod(0,5,7)$

NB: Please draw horizontally on your script.

NB: You can't simplify the above functions and then draw using NAND gates. You have to draw based on the function given in question

4. Draw the following functions using NOR gates only:

- a. $F(A,B,C) = A' + A'C'$
- b. $F(A,B,C,D) = (AB'C'D' + AD + (B+D'))$
- c. $F(w,x,y,z) = \sum(0,1,2,3,9,11)$

NB: Please draw horizontally on your script.

NB: You can't simplify the above functions and then draw using NOR gates. You have to draw based on the function given in question

5. **Convert** the following equations to their canonical SOP and POS form using Boolean Algebra:

- a. $F(A,B,C) = AB + BC'$
- b. $F(A,B,C,D) = A + B'CD'$